

Super-Resolution Reconstruction of Turbulent Flows with Machine Learning

Reconstructed no-skill baseline

Abstract

This paper presents a deep learning framework for super-resolution reconstruction of turbulent flows. A convolutional neural network accurately reconstructs high-resolution velocity and vorticity fields from low-resolution measurements. The method is validated on a cylinder wake at $Re_D = 100$ and homogeneous isotropic turbulence, where it outperforms bicubic interpolation and achieves robust recovery of small-scale structures. The results demonstrate that machine learning can provide an efficient and generalizable approach for revealing turbulent structures from coarse CFD and experimental data.

1 Introduction

High-resolution flow fields are essential for understanding turbulence, wake dynamics, and coherent structures. However, direct numerical simulation and experimental measurement are expensive and often produce under-resolved data. Super-resolution offers a way to recover fine-scale flow information from coarse observations. Inspired by computer vision, this paper uses convolutional neural networks to learn the mapping from low-resolution flow images to high-resolution flow states.

The main contribution is a robust machine-learning method that reconstructs turbulent flow fields with high accuracy and low training-data requirements. Unlike classical interpolation, the proposed model learns nonlinear flow features and therefore recovers small vortical structures and energy-containing scales. This capability is important for real-time flow diagnostics and data-driven CFD acceleration.

2 Method

Let $\mathbf{q}_{LR}$ be the low-resolution input and $\mathbf{q}_{HR}$ be the high-resolution reference field. The neural network $\mathcal{N}_\theta$ is trained to minimize

$$\mathcal{L}(\theta) = \|\mathcal{N}_\theta(\mathbf{q}_{LR}) - \mathbf{q}_{HR}\|_2^2. \quad (1)$$

The architecture consists of convolutional layers with skip connections and multi-scale feature extraction. Average and max pooling generate low-resolution fields, while the output layer reconstructs high-resolution velocity and vorticity.

The training data are generated by DNS of incompressible Navier–Stokes flows. The cylinder wake is simulated at $Re_D = 100$, and a two-dimensional homogeneous turbulence case is used to test high-wavenumber reconstruction. The data are split into training, validation, and test sets. Adam optimization and early stopping are used.

3 Experiments

The first experiment reconstructs the cylinder wake. Figure 1 compares ground truth, low-resolution input, bicubic interpolation, and the neural prediction. The proposed model reconstructs the vortex street and reduces L_2 error relative to interpolation. The method also performs well with only 50 training snapshots.

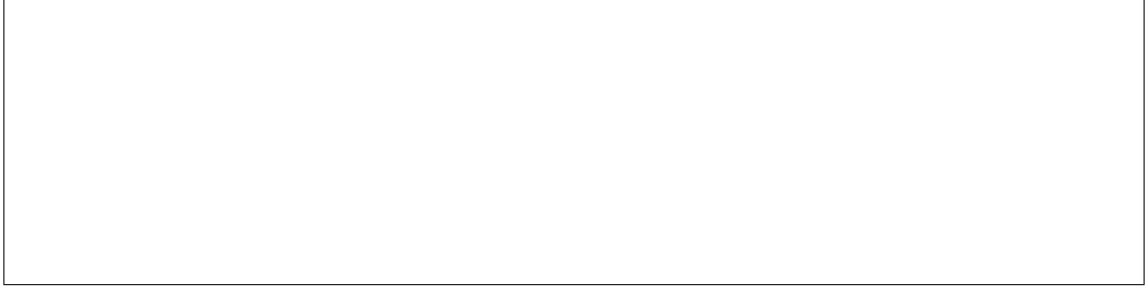


Figure 1: Cylinder wake reconstruction at $Re_D = 100$. Ground truth, coarse input, interpolation, and neural reconstruction are compared.

The second experiment reconstructs two-dimensional homogeneous turbulence. Figure 2 shows that the network recovers coherent vortical structures and small-scale fluctuations. Spectral comparisons demonstrate that the model accurately restores high-wavenumber content better than bicubic interpolation.

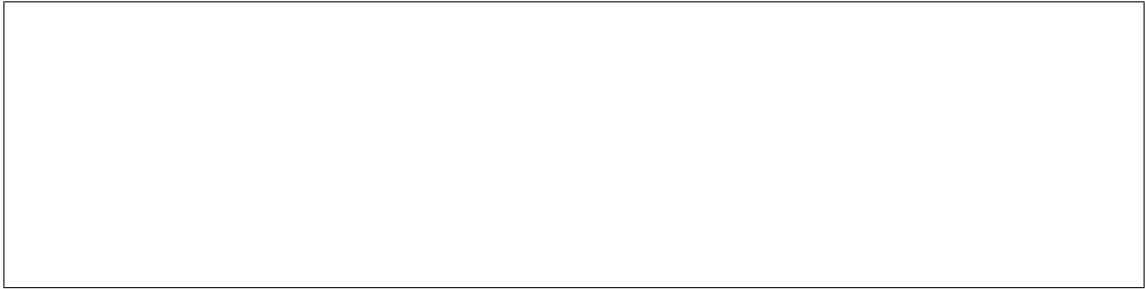


Figure 2: Homogeneous turbulence reconstruction. The machine-learning model recovers fine vortical structures and high-wavenumber energy.

Case	Baseline	Result
Cylinder wake	Bicubic	Lower error and better PDF agreement
Homogeneous turbulence	Bicubic	Better small-scale recovery
Limited data	CNN	Accurate with 50 snapshots

Table 1: Summary of reconstruction results.

4 Discussion

The experiments show that convolutional networks can learn physically meaningful reconstruction operators for fluid data. The method is robust across laminar and turbulent cases and provides a

promising route toward real-time reconstruction of under-resolved simulations and measurements. Because the model is data-driven, accuracy can improve further with larger datasets and more advanced architectures.

5 Conclusion

Machine-learning super-resolution can reconstruct high-resolution flow fields from coarse measurements. The method outperforms bicubic interpolation, captures turbulent structures, and works with limited training data. These results indicate that deep learning is a powerful and general tool for turbulent-flow reconstruction.

References

- [1] Fukami, Fukagata, and Taira. Super-resolution reconstruction of turbulent flows with machine learning. *Journal of Fluid Mechanics*, 2019.